

The Effect of Review Summary: Evidence from a Natural Experiment

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Abstract

By leveraging a unique policy of an online review platform that introduces generative AI reviews summary (GAIRS), this study examines how GAIRS affect the characteristics of user-generated content in the online reviews context. Constructing a unique dataset of online reviews for a matched set of hotels across TripAdvisor and Expedia, we apply a cross-platform difference-in-differences approach to assess the impact of GAIRS. Our findings elucidate the adverse effects of GAIRS on users' subsequent contributions, manifested in diminished quantity and length of subsequent reviews. Nonetheless, GAIRS also correlates with an increase in the average ratings of reviews. We identify the substitution and learning effects as two plausible explanations for these effects. Further analyses reveal that the substitution effect mainly reduces strongly negative reviews, particularly in lower-status hotels, while the learning effect primarily causes inexperienced users to emulate GAIRS. This research contributes to the expanding discourse on how GAI impacts user-generated content.

Keywords: generative AI reviews summary, user-generated content, natural experiment, difference-in-differences, substitution and learning effects

1. Introduction

Online review platforms are crucial for consumers, offering rich review information that reduces purchase uncertainty (Wu et al., 2015; Zhu & Zhang, 2010). Consumers can utilize both quantitative information like valence and volume (Chintagunta et al., 2010) and qualitative content of reviews (Chakraborty et al., 2022) to assess product quality and suitability. However, such abundant information can overwhelm users, leading to cognitive overload (Davis et al., 2021), especially when encountering conflicting signals (Branco et al., 2016). Recent advancements in generative artificial intelligence (GAI) have shown promising potentials in addressing this challenge. Several online platforms,

such as Amazon and TripAdvisor, have recently started to integrate GAI-generated content into their systems.

Trained on large foundation models with extensive datasets, GAI has the potential to streamline the presentation of information by efficiently filtering and summarizing content. However, concerns persist regarding the integration of GAI-generated content, particularly in terms of veracity, reliability, potential violations of intellectual property rights (Susarla et al., 2023), and the possible detrimental effects on individuals' creative abilities (Burtch et al., 2023). Consequently, while GAI-generated content may enhance information acquisition, it is important to carefully examine the impacts of GAI-generated content, particularly its influence on user-generated content (UGC) dynamics on online platforms.

Some studies have explored the role of GAI across various platforms, mainly focusing on scenarios where GAI provides external knowledge (Burtch et al., 2023; Shan & Qiu, 2023; Xue et al., 2023) or serves as platform-integrated content generation tools (Huang et al., 2023; Zhang et al., 2023). Our study differs from prior research by studying a more direct and transparent incorporation of GAI, specifically through prominently displaying GAI review summaries (GAIRS) on the top of review webpage (see Figure 1 within the context of TripAdvisor). Notably, GAIRS, summarized from recent UGC, provides a comprehensive overview from both positive and negative aspects and a set of quality attributes of an organization. Furthermore, the generation process of GAIRS in our study is transparent, with clear traceability to the original source of user reviews (see Figure 2). This transparency contrasts with other GAI-generated contents, which often lack such clarity (Lebovitz et al., 2022). Consequently, the impact of GAI review summaries on the dynamics of subsequent user content generation remains an unexplored question in this unique context.

Understanding this question is important because it can help platforms shape more informed policies and better implement and manage their GAI function. Theoretically, there are competing aspects regarding whether online platforms should incorporate GAI-generated content. In the incidence-decision stage (i.e.,

whether to contribute, Moe & Schweidel, 2012), owing to the superior performance of GAI compared to humans (Susarla et al., 2023), GAI-generated content may augment the content of online platforms by providing fresh discussion points and often-overlooked aspects (Lu & Zhang, 2024) that inspire users, thereby stimulating the subsequent contribution (Shan & Qiu, 2023; Su et al., 2023). Conversely, given that GAI-generated content is easily accessible and comprehensive, it also may unintentionally discourage users by triggering substitution effect (Lysyakov & Viswanathan, 2023) or by crowding out intrinsic motivation, thereby diminishing the subsequent user contribution (Shan & Qiu, 2023; Su et al., 2023).



Figure 1. GAI: Sample from TripAdvisor.

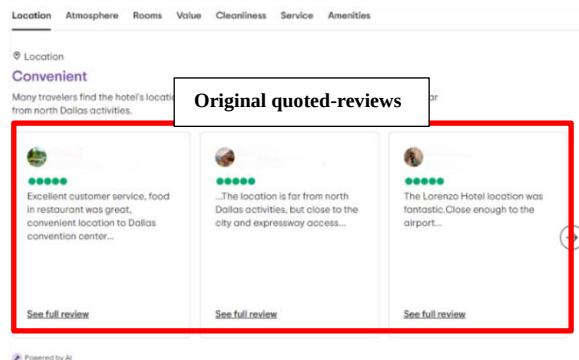


Figure 2. Transparency of GAI: Traceable reviews.

In the evaluation-decision stage (i.e., how to contribute), previous studies have revealed that users can be influenced by professional content (e.g., generated by experts), exhibiting herding behavior (Deng et al., 2022; Jacobsen, 2015). Along this line, GAI may also induce users learning, as it offers a comprehensive summary of previous reviews and captures a broad spectrum of user opinions. Consequently, users may produce content resembling GAI. On the other hand, users may question the reliability of GAI, or may provide independent analyses reflecting personal experiences and opinions,

as suggested by Berger (2014). This may prompt users to offer content distinct from GAI, thereby presenting a differentiation effect. Considering the research gap and competing effects of GAI on users' contribution behavior, we propose the following research questions: (1) *Does the introduction of GAI exert an impact on the subsequent users' review behavior? If so, how does it affect the quantity and quality of users' subsequent reviews?* (2) *What is the underlying mechanism of such impact?*

We empirically study these research questions by exploiting an exogenous event on TripAdvisor. We perform a causal identification by using a cross-platform difference-in-differences (DID) approach. Our empirical analyses yield several important findings. First, following the launch of GAI, hotels on TripAdvisor experienced a significant decrease in the total number of subsequent user reviews, suggesting that GAI's substitution effect may outweigh its inspiration effect on user reviews. Moreover, hotels on TripAdvisor saw a significant increase in review rating and a reduction in review length. Due to GAI's inherent positive sentiment bias (e.g., higher rating) and lack of concreteness (e.g., shorter length), our results may suggest that users who produce subsequent reviews may exhibit a learning effect towards GAI in the evaluation-decision stage.

Second, we further validate the plausibility of the substitution and learning effects as underlying mechanisms for the main results. Our analysis indicates that after the introduction of GAI, experienced users significantly reduce their contribution in reviews. Additionally, the number of organic reviews (versus incentivized reviews) also decline, suggesting a reduced user motivation in writing regular, non-incentivized reviews. Moreover, the reduction in review volume is more pronounced when the perceived value of GAI is high, i.e., when GAI is generated based on a larger number of previous user reviews, encompasses a greater number of attributes, or exhibits greater readability. In the evaluation-decision stage, we note that after the launch of GAI, users' subsequent reviews are more similar to AIG in terms of topic, sentiment, length, and readability. The results provide empirical evidence for the learning effect, suggesting that users who are willing to contribute may learn from the GAI characteristics when writing regular reviews. In addition, due to GAI's inherent characteristics, the subsequent user ratings tend to be higher, and review length be shorter. All these findings provide additional insights into our mechanism.

Third, our further analyses reveal that the decrease in review volume due to the substitution effect stems primarily from a reduction in strongly negative reviews, and this trend is particularly noticeable among lower-

status hotels. Furthermore, the similarity between user reviews and GAIRS resulting from the learning effect occurs primarily among inexperienced users who may emulate GAIRS.

Our study makes several contributions. First, we contribute to expanding literature on the impact of GAI on UGC. While previous research has mainly examined contexts where GAI functions as either external sources of knowledge provision (Burtch et al., 2023; Shan & Qiu, 2023; Xue et al., 2023) or platform-integrated tools for content generation (Zhang et al., 2023), our study investigates a more direct and transparent application of GAI. Specifically, we analyze the effects of prominently displaying GAIRS, with clear traceability to the original source, on the subsequent user reviews. The novel form of GAIRS allows us to explore the effects of GAI-generated content more directly and provide richer insights into the underlying mechanism.

Second, although the impact of professional reviews on the subsequent reviews has been well studied (e.g., Deng et al., 2022; Jabr & Rahman, 2022), these findings may not be directly applicable to analyzing the implication of GAIRS. This is due to the substantial differences between human-generated content and GAI-generated content. Furthermore, existing research has shown that consumer react to two different types of content differently (Dennis et al., 2023; Kim et al., 2022). Our research contributes to this literature by studying the distinct impact of GAIRS on the subsequent reviews. We further demonstrate the substitution effect and learning effect as the underlying mechanisms. The theoretical development and empirical analyses of the mechanism enrich the understanding of the review-generating behaviors. Finally, considering the inherent challenges online platforms face in facilitating and managing user contribution behavior, our study offers practical insights on consumer interactions with GAI-generated content.

2. Literature Review

2.1. GAI and user contribution behaviors on platforms

GAI refers to a category of AI models designed to generate novel content, including text, images, and other forms of media (Susarla et al., 2023). Given its remarkable content-creation capabilities, GAI has garnered significant attention from researchers. A prominent topic of discussion is the impact of GAI on user contribution behaviors on platforms. Existing research has primarily focused on two types of GAI: platform-external and platform-integrated GAI tools. For platform-external GAI tools, such as ChatGPT, studies have shown that it can lead to an increase in the

volume of content contribution (Burtch et al., 2023; Shan & Qiu, 2023). However, findings related to changes in content length and quality are inconsistent. Some studies report longer and more complex content contribution (Sanatizadeh et al., 2023), while others observe shorter content contribution (Burtch et al., 2023; Shan & Qiu, 2023). For the latter GAI tools, the findings are varied. Huang et al. (2023) report that the deployment of a GAI tool on the Lofter platform led to a decrease in content creators' activities. In contrast, Zhang et al. (2023) find that the adoption of AI voice technology on TikTok enhances creator productivity and increases the novelty of videos. Although this stream of research has advanced our understanding of the value of GAI adoption, no research has directly investigated the effects of direct and visible application of AI-generated content, particularly with regard to its traceability. The present study addresses this void.

2.2. Impact of professional reviews on review generation

Our work also aligns with and extends an emerging body of research that investigates the influence of professional reviews within the context of online reviews. The existing literature has documented the significant impacts of various types of professional reviews on users' contribution behavior. For instance, Deng et al. (2022) demonstrate that editorial reviews can profoundly influence subsequent user review behavior, affecting aspects such as review ratings, volume, and sentiment. Similarly, Jacobsen (2015) finds that consumer ratings tend to align with expert reviews, with this alignment effect being particularly pronounced in response to strongly negative or strongly positive expert opinions. Jabr and Rahman (2022) show that professional evaluations can reduce users' difficulties in assessing information, thereby alleviating information overload. Further, Zhang et al. (2016) reveal that an increase in the number of expert reviews correlates with higher future traveler ratings, though the marginal effect diminishes.

In contrast to these studies, our research explores a novel category of "professional reviews": those generated by non-human entities, specifically GAI. Given the substantial differences between human-generated and machine-generated content, our study provides a unique contribution to the literature on top reviews. We not only investigate the distinct characteristics and impacts of GAIRS but also elucidate the underlying mechanisms driving these effects. By doing so, our research offers valuable practical implications for online review platforms, enhancing their understanding of how to effectively integrate GAI-generated content to optimize user engagement.

3. Research Methodology

3.1. Research context

Our research setting is TripAdvisor, one of the largest online platforms for accommodations, restaurants, and attractions in the world (Gao et al., 2024). TripAdvisor has enormous volume of user-generated reviews, which not only helps millions of travelers book their trips but also benefits businesses worldwide. Despite the substantial benefits of online reviews, the sheer abundance of information can be overwhelming, leading to cognitive overload for consumers (Davis et al., 2021). To address this challenge, TripAdvisor introduced AIGRS for some hotels in November 2023. This innovation leverages GAI to create concise summaries of recent user reviews, thereby helping users quickly grasp essential insights from reviews. Notably, the selection of hotels that display AIGRS is determined by the platform, rather than by the individual hotels. The AIGRS are prominently displayed on the top of review pages of selected hotels, accompanied by a disclosure clause labeled "*Traveler Insights*" at the beginning of each summary (see in Figure 1). In addition, these summaries also highlight a set of key attributes of the hotels, along with a one-word opinion (e.g., Location: Convenient). Each attribute includes detailed quoted-information (e.g., review valance and text) from the original user reviews used to generate the summary (see in Figure 2).

3.2. Data and variables

Our data sample comprises hotels located in Texas, California, and New York, sourced from two prominent platforms: TripAdvisor and Expedia. Initially, we collected all user reviews posted until February 2024 for 619 hotels listed on TripAdvisor that were treated with GAIRS. For each review, we recorded the reviewer ID, submission date, star rating, and text content. Additionally, we gathered detailed hotel-level information, including specifics about GAIRS, hotel rank, type, and star-class. Our second data source, Expedia, is another widely-used travel platform that hosts a vast array of user-generated reviews. For the same 619 hotels matched from TripAdvisor, we collected all user reviews available on Expedia, recording similar information for each review, except for GAIRS details, which Expedia does not feature. We aggregate the review data to the hotel-platform-month level and focus our analysis on the three months before and after the launch of GAIRS on TripAdvisor, resulting in a final dataset comprising 8,666 observations. As the core variable of our study, GAIRS characteristics are

presented statistically in Table 1, which shows a strong positive sentiment (e.g., sentiment of GAIRS). Figure 3 further illustrates this positive bias.

Table 1. Definition and statistics of GAIRS characteristic variables.

Variable	Mean	SD	Min	Max
<i>Num_Reference</i>	65.02	45.45	20	487
<i>Num_Attribute</i>	7.00	0.51	5	8
<i>Readability_GAIRS</i>	13.96	1.17	10.7	17.5
<i>Length_GAIRS</i>	613.79	66.43	470	919
<i>Sentiment_GAIRS</i>	0.87	0.27	-0.96	0.99

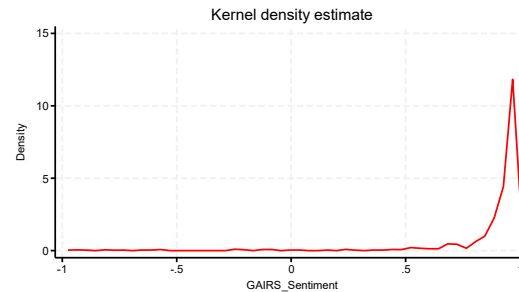


Figure 3. Distribution of GRARS sentiment.

3.3. Empirical model

When identifying the potential effects of GAIRS, it is important to note that on the same platform, hotels that selected to have GAIRS may be systematically different from hotels that are not selected. Therefore, we utilize the same hotels on a different platform (Expedia) that does not offer GAIRS as the control group. Intuitively, GAIRS should affect the reviews of hotels on TripAdvisor but not those of the same hotels on Expedia. Specifically, we use a cross-platform DID identification strategy, which compares the changes in a hotel's reviews on TripAdvisor following the launch of GAIRS with changes in the same hotel's reviews on Expedia over the same time period. This strategy allows us to control for the unobserved shocks to hotel reviews (e.g., improvements in service quality) that are common to both platforms. We analyze the change in user review behavior after the launch of GAIRS using the following model specification:

$$DV_{ipt} = \beta TripAdvisor_p + \delta TripAdvisor_p \times After_{it} + \gamma X_{ipt} + \alpha_i + \tau_t + \varepsilon_{ipt} \quad (1)$$

where DV_{ipt} represents different dependent variables: the log volume of reviews, the average rating of reviews, and the log average length of reviews for hotel i in month t on platform p , respectively. The variable $After_{it}$ is a binary indicator set to one if month t is during or after the launch of GAIRS, and zero otherwise. The variable $TripAdvisor_p$ indicates whether the reviews

are sourced from TripAdvisor. The variable X_{ipt} represents control variables that characterize the review environment on each platform in month t . Specifically, we control for the number, average rating, and average length of previous reviews, following the methodology of Proserpio and Zervas (2017). To control for unobserved heterogeneity, we include a matched-pair fixed effect α_i , which accounts for shared characteristics of the same hotel i across different platforms. Additionally, we incorporate year-month fixed effects τ_t to control the influence of temporal shocks that affect hotels at a given month t . The error term ε_{ipt} captures idiosyncratic random disturbances. We cluster all standard errors at the hotel level to account for the potential correlation of error terms within a hotel. In this specification, the coefficient δ represents the DID estimate of interest, quantifying the impact of GAIRS on the review volume, average rating, and average review length, respectively.

4. Empirical Analysis and Results

4.1. Main results

We present the examination results of the DID analyses in Table 2. Column 1 of Table 2 reports the results for review volume. The estimated coefficient for the interaction term *TripAdvisor* $\times$ *After* is negative and statistically significant, indicating that the launch of GAIRS on TripAdvisor leads to a 17.96% decrease in the number of reviews per month. This finding suggests that GAIRS may discourage the subsequent review generation. Columns 2 and 3 of Table 2 present the results for the average rating and the average length of reviews, respectively. Specifically, the launch of GAIRS results in a statistically significant increase in the average rating of reviews by 0.056 points. Conversely, the introduction of GAIRS is associated with a 3.92% decrease in the average length of subsequent reviews. Given that GAIRS's inherent positive sentiment bias and lack of concreteness, our findings provide initial evidence that users who are

willing to contribute in writing regular reviews may exhibit a learning effect from GAIRS. These results collectively indicate that GAIRS has a notable impact on the characteristics of the subsequent online reviews.

Table 2. Effect of GAIRS on aggregated review characteristics.

	(1) <i>log(Volume)</i>	(2) <i>Avg. Rating</i>	(3) <i>log(Avg. Length)</i>
<i>TripAdvisor</i> $\times$ <i>After</i>	-0.198*** (0.025)	0.056** (0.025)	-0.040* (0.021)
<i>TripAdvisor</i>	-0.718*** (0.043)	-0.166*** (0.025)	1.128*** (0.018)
Time FE	YES	YES	YES
Hotel FE	YES	YES	YES
Controls	YES	YES	YES
Observations	8,666	8,281	8,208
R-square	0.604	0.421	0.625

Notes. Clustered robust standard errors at the hotel level are in parentheses. FE, fixed effect. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The same notes apply to Tables 3–12.

4.2. Robustness checks

We first employ a relative time model, following the methodology of Angrist and Pischke (2009), to assess the validity of the parallel trend assumption underpinning our identification strategy. We use the one-month interval before the launch of GAIRS as the baseline period. Figure 4 presents visualizations of our results. The observed patterns in these dynamic trends lend further support to the validity of our results. Second, we further include city by year-month fixed effects to control for the time-varying regional and macro factors that can potentially influence our results and reestimate our model. Third, we exclude hotels exhibit an average of fewer than one review per month over a seven-month period on either of the two platforms to reduce potential estimation bias. Finally, we further exclusively focus on English reviews to estimate our model, as GAIRS is generated from recent English reviews and is likely to primarily affect subsequent English reviews. All the corresponding results consistently support the validity of our identification strategy.

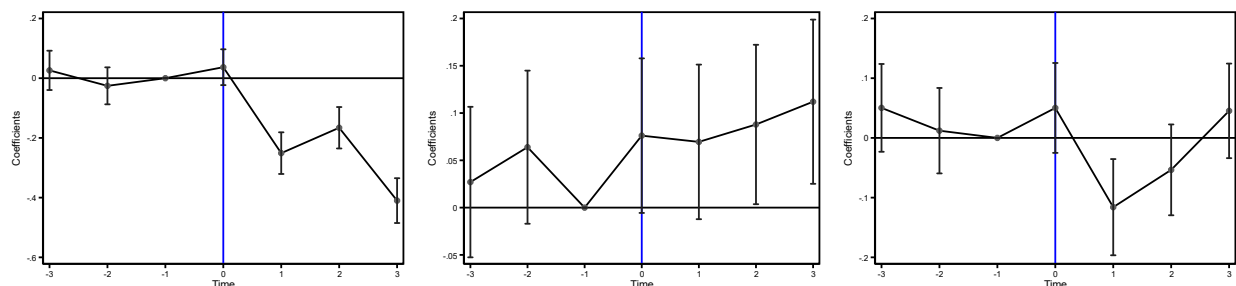


Figure 4. Parallel trend test for review volume, average rating, and average review length.

5. Mechanism Analysis

5.1. Substitution effect: Experienced vs Inexperienced users, Organic vs Solicited reviews

To further investigate the mechanism on the decline in the number of reviews, we examine whether the intrinsic motivation diminishes due to the substitution effect. According to literature, we focus on individuals with different levels of motivation, characterized by their experience and the source of reviews (Su et al., 2023; Gao et al., 2024). Given that experienced individuals and those providing original reviews may have higher motivation in contributing content, we expect the negative impact of GAIRS introduction on the subsequent review volume to be more salient for

users who have higher intrinsic motivation. In our study, individuals are categorized as experienced if their review contributions are above the median, and inexperienced if below. Source is determined by whether a review is organic (regular) or solicited (incentivized).

We then construct a matched dataset using propensity score matching (PSM) to analyze the changes in review volume across different motivation groups within the matched hotels. The results are presented in Table 3. The results reveal a more significant decrease in the number of reviews from experienced users compared to inexperienced users, resulting in a reduced proportion of reviews from experienced users. Furthermore, we find a similar results regarding organic reviews, which indicate the decline of user motivation. These findings align with the perspective of substitution effect, suggesting a weakening of intrinsic motivation.

Table 3. Substitution effect: GAIRS's influence on review volumes across different motivation groups.

	(1) <i>Num_Exp</i>	(2) <i>Num_InExp</i>	(3) <i>Ratio_Exp</i>	(4) <i>Num_Org</i>	(5) <i>Num_Sol</i>	(6) <i>Ratio_Org</i>
<i>Treat × After</i>	-0.118*** (0.026)	-0.094*** (0.031)	-0.039** (0.020)	-0.118*** (0.031)	-0.077*** (0.022)	-0.008** (0.004)
Hotel/Time FE	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES
Observations	5,859	5,859	4,401	5,859	5,859	4,401
R-square	0.714	0.787	0.411	0.784	0.866	0.808

5.2. Substitution effect: Interpretability of GAIRS

Consumers' perception of AI value can significantly influence their behavior. Prior research indicates that the quality of AI output is a key factor in shaping this perception (Hou et al., 2023; Liang et al., 2024), particularly through its interpretability (Bauer et al., 2023). In our study, given that GAIRS consists of mainly text information, its readability is a critical aspect influencing interpretability. Higher readability facilitates users in extracting information from GAIRS, thereby shaping their positive perception towards the interpretability of GAIRS. In addition, since GAIRS is derived from recent reviews, the volume of reference reviews may contribute to its interpretability. A larger volume of reference reviews used in generating GAIRS may enhance its representativeness and interpretability. Conversely, a small volume of reference reviews may have limited ability to capture diverse opinions, thus reducing the interpretability.

Building upon the discussion above, we anticipate that GAIRS with higher interpretability will result in a greater substitution effect due to their enhanced utility. Initially, we use the SMOG formula to determine the years of education needed to understand the review text, following Ghose and Ipeirotis (2011). By using the number of polysyllabic words, we calculate the SMOG score for each GAIRS and then categorize them into two groups: high readability and low readability. We further distinguish between GAIRS derived from a high volume of reference reviews and those from a low volume and then conduct subgroup analyses. The findings in Table 4 reveal that the interpretability of GAIRS influences user review behavior. Specifically, high readability and GAIRS derived from a large number of reference reviews exhibit a greater impact on review volume. These findings provide further support for the substitution effect of GAIRS. We also utilize GAIRS length and the number of attributes to measure interpretability, providing further validation of this mechanism.

Table 4. Impact of GAIRS's readability and reference review volume.

<i>log(Volume)</i>	(1) <i>High Readability</i>	(2) <i>Low Readability</i>	(3) <i>Large Reference Number</i>	(4) <i>Small Reference Number</i>
<i>TripAdvisor × After</i>	-0.241*** (0.033)	-0.148*** (0.038)	-0.247*** (0.034)	-0.138*** (0.037)
TripAdvisor	YES	YES	YES	YES
Hotel/Time FE	YES	YES	YES	YES
Controls	YES	YES	YES	YES
Observations	4,634	4,032	4,774	3,892
R-square	0.612	0.594	0.567	0.572

5.3. Substitution effect: Characteristics of hotel

We now investigate how GAIRS influences users' review behavior, considering the hotel's star-class, quality, and rank. A stronger substitution effect is expected in lower-status hotels due to higher information asymmetry compared to higher-status hotels. Subgroup analyses results are presented in Table 5, showing that GAIRS has a more pronounced impact on review volume in low-status (below 4 stars), low-quality (below median hotel rating), and low-rank (above median ranking) hotels. These findings provide additional support for our mechanisms.

5.4. Learning effect: Similarity between user reviews and GAIRS

To test the learning effect in the evaluation-decision stage, we examine changes in user reviews after the launch of GAIRS. Following Deng et al. (2022), we conduct review-level analyses focusing on content topic, sentiment, length, and readability to identify differences between user reviews and GAIRS. For content topic, we use latent Dirichlet allocation (LDA) (Blei et al., 2003) to uncover hidden semantic structures in the reviews. We then compare the similarity between user reviews and GAIRS by calculating the cosine similarity of their topic vectors (Deng et al., 2022). Regarding sentiment, we employ a rule-based approach using the Valence Aware Dictionary and Sentiment Reasoner (VADER), a sentiment analysis tool designed for social media text (Hutto & Gilbert, 2014). This method assigns positive,

negative, and neutral sentiment scores to each review and GAIRS. We again employ SMOG score to assess readability. Then we focus on the absolute differences in sentiment score, content length, and SMOG score to determine their corresponding similarities between user reviews and GAIRS.

The results of the review-level analysis are presented in Table 6, indicating that the subsequent reviews become more similar to GAIRS after its launch across all examined measures. These results suggest that users are influenced by AGIRS, leading them to form similar reviews. Due to GAIRS's inherent positive sentiment bias and lack of concreteness, user ratings tended to be higher, and review lengths shorter. We further explore whether user reviews actually become less concrete after the launch of GAIRS. We confirm this by examining the impact of GAIRS on the concreteness of words in user reviews following the method of Pan et al. (2018). As indicated in the Table 7, following the launch of GAIRS, the concreteness of user reviews has notably declined. This suggests that user reviews have become more general and abstract, resulting in a reduction in review length.

Table 7. The effect of GAIRS on review concreteness.

	<i>Concreteness</i>
<i>TripAdvisor × After</i>	-0.203** (0.099)
TripAdvisor	YES
Hotel/Time FE	YES
Controls	YES
Observations	8,281
R-square	0.133

Table 5. Impact of different hotel status.

<i>log(Volume)</i>	(1) <i>Low-class</i>	(2) <i>High-class</i>	(3) <i>Low-quality</i>	(4) <i>High-quality</i>	(5) <i>Low-rank</i>	(6) <i>High-rank</i>
<i>TripAdvisor × After</i>	-0.252*** (0.038)	-0.132*** (0.035)	-0.219*** (0.038)	-0.176*** (0.032)	-0.209*** (0.033)	-0.182*** (0.038)
TripAdvisor	YES	YES	YES	YES	YES	YES
Hotel/Time FE	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES
Observations	4,410	3,822	4,354	4,312	5,124	3,542
R-square	0.624	0.615	0.642	0.562	0.639	0.565

Table 6. Learning effect: Similarity between user reviews and GAIRS.

	(1) $\Delta Topic$	(2) $\Delta Positive$	(3) $\Delta Neutral$	(4) $\Delta Negative$	(5) $\Delta Length$	(6) $\Delta SMOG$
GAIRS	0.007*** (0.002)	-0.002* (0.001)	-0.003*** (0.001)	-0.001*** (0.000)	-12.410** (5.093)	-0.127*** (0.037)
Hotel/Time FE	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES
Observations	89,349	89,349	89,349	89,349	89,349	89,349
R-square	0.113	0.078	0.088	0.202	0.054	0.177

6. Additional Analysis

We have verified two potential mechanisms of GAIRS's impact on user review behavior, namely the substitution effect and the learning effect. However, questions remain about which type of reviews are reduced by the substitution effect and whether this impact varies among different hotels. Additionally, we need to determine who contributes to the content similarity caused by the learning effect. To address these questions, we conducted further research, with the results presented in Tables 8 to 12 in the appendix. We found that the decrease in review volume due to the substitution effect mainly results from a reduction in strongly negative reviews, particularly among lower-status hotels. This offers a new perspective on the decline in review quantity and the rise in ratings. Additionally, the similarity between user reviews and GAIRS due to the learning effect primarily occurs among inexperienced users who emulate GAIRS.

Table 8. GAIRS's influence on the ratio of different review.

	(1) <i>Ratio_Pos</i>	(2) <i>Ratio_Neu</i>	(3) <i>Ratio_Neg</i>
<i>TripAdvisor × After</i>	0.005 (0.005)	-0.001 (0.004)	-0.012** (0.005)
TripAdvisor	YES	YES	YES
Hotel/Time FE	YES	YES	YES
Controls	YES	YES	YES
Observations	8,281	8,281	8,281
R-square	0.380	0.173	0.368

7. Discussion and Conclusion

As the volume of online reviews continues to grow, consumers are increasingly confronted with information overload. This phenomenon can make it challenging for users to effectively process and utilize the abundance of available information. As such, review platforms are adopting GAI to address this challenge. In this paper, we study the effect of GAIRS launched by TripAdvisor on subsequent review behavior. By applying a cross-platform DID model on reviews posted on TripAdvisor and Expedia, we find that GAIRS leads to a reduction in the total number of reviews and the length of reviews, while increasing overall review ratings. We attribute these findings to the substitution effect and learning effect.

These findings offer valuable insights for review platform owners. The inherent nature of GAI suggests that platform-initiated GAIRS could effectively address information overload issues. However, the observed decline in review volume and length underscores the need for careful consideration and thorough evaluation of potential benefits and drawbacks before implementation. Platforms should consider the specific characteristics and likely reactions of their target businesses and users. Should platforms decide to adopt GAIRS, they must develop strategies to mitigate potential adverse impacts and ensure that the information provided by GAIRS is comprehensive and interpretable, as these factors significantly influence user perception and review dynamics.

Table 9. GAIRS's influence on the ratio of different review (Based on hotel star class).

	(1)	(2)	(3)	(4)	(5)	(6)
	Low-class Hotel			High-class Hotel		
	<i>Ratio_Pos</i>	<i>Ratio_Neu</i>	<i>Ratio_Neg</i>	<i>Ratio_Pos</i>	<i>Ratio_Neu</i>	<i>Ratio_Neg</i>
<i>TripAdvisor × After</i>	0.010 (0.009)	0.001 (0.006)	-0.021*** (0.008)	-0.002 (0.007)	0.001 (0.006)	-0.004 (0.007)
TripAdvisor	YES	YES	YES	YES	YES	YES
Hotel/Time FE	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES
Observations	4,172	4,172	4,172	3,728	3,728	3,728
R-square	0.422	0.184	0.417	0.285	0.154	0.279

Table 10. GAIRS's influence on the ratio of different review (Based on hotel quality).

	(1)	(2)	(3)	(4)	(5)	(6)
	Low-quality Hotel			High-quality Hotel		
	Ratio_Pos	Ratio_Neu	Ratio_Neg	Ratio_Pos	Ratio_Neu	Ratio_Neg
TripAdvisor	0.009	0.000	-0.021**	0.000	-0.002	-0.003
× After	(0.008)	(0.007)	(0.008)	(0.007)	(0.005)	(0.007)
TripAdvisor	YES	YES	YES	YES	YES	YES
Hotel/Time FE	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES
Observations	4,157	4,157	4,157	4,124	4,124	4,124
R-square	0.405	0.177	0.394	0.287	0.145	0.271

Table 11. GAIRS's influence on the ratio of different review (Based on hotel rank).

	(1)	(2)	(3)	(4)	(5)	(6)
	Low-rank Hotel			High-rank Hotel		
	Ratio_Pos	Ratio_Neu	Ratio_Neg	Ratio_Pos	Ratio_Neu	Ratio_Neg
TripAdvisor	0.010	-0.002	-0.020***	-0.003	0.000	0.001
× After	(0.008)	(0.006)	(0.008)	(0.006)	(0.005)	(0.006)
TripAdvisor	YES	YES	YES	YES	YES	YES
Hotel/Time FE	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES
Observations	4,859	4,859	4,859	3,422	3,422	3,422
R-square	0.390	0.152	0.384	0.240	0.170	0.223

Table 12. Who leads to the similarity between user reviews and GAIRS.

	(1)	(2)
	Inexperienced Users Reviews	Experienced Users Reviews
	$\Delta Topic$	$\Delta Topic$
GAIRS	0.006** (0.003)	0.005 (0.004)
Hotel/Time FE	YES	YES
Controls	YES	YES
Observations	51,894	30,422
R-square	0.121	0.149

Our findings also have important implications for business managers who are facing the launch of GAIRS. We show that GAIRS is likely to benefit firms in review valence, but the effect on review valence is heterogeneous across businesses of different status. Specifically, popular businesses with high status do not experience the same increase in review valence as less popular ones. Therefore, there is need for these businesses to adopt the practice to solve this challenge. For example, they can gain insights into their own shortcomings from GAIRS and subsequently make targeted improvements in these areas. Conversely, less popular hotels benefit from higher ratings facilitated by GAIRS but must also contend with a decrease in review quantity and length. These businesses need to carefully balance the benefits and potential challenges associated with GAIRS, adopting strategies to address possible drawbacks.

8. References

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